NeST-Com: Fixed-Budget Next-Scale Token Communication

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Abstract

NeST-Com is a digital image communication framework that uses next-scale residual descriptions under fixed per-image symbol budgets. It coordinates source-prefix selection, channel protection and prefix-constrained reconstruction through actual-length frame fitting. It retains scale-wise probability evaluation while fitting independently terminated within-scale prefixes; received boundary tokens refine the residual state and finer-scale predictions. At 1024 symbols and 10 dB, source-bit capacity utilization increases from 77.70% to 99.79%, reducing DISTS and DreamSim by 8.7% and 20.5% relative to shared-model whole-scale transmission. Matched controls and system-level comparisons characterize reconstruction gains and quality–cost tradeoffs.

Index Terms

Image communication, next-scale tokens, fixed symbol budget, perceptual reconstruction.

I. Introduction

Next-scale representations describe an image through complete coarse maps followed by residual refinements [1]. Each scale map spans the full spatial extent of the latent field and provides context for finer-scale probability prediction. For communication, this organization offers both a progressively refinable source description and context for reconstructing untransmitted content.

ARPC enables next-scale progressive image compression through scale selection, entropy coding, generation and adaptive quantization [2]. Ada-TokenCom jointly adapts next-token prefix length and modulation and coding scheme (MCS) under a long-term symbol budget, using the received prefix to generate the remaining tokens [3]. ResARC further codes residuals to compensate for suffix-generation errors [4]. These studies establish foundations for progressively transmitting and reconstructing image information through source coding, prefix transmission and generation-error compensation.

For communication with fixed-length digital frames per image, source-code termination must account for the terminated arithmetic-code length, signaling overhead and channel protection [5]. When the next complete residual map cannot fit, whole-scale transmission can leave capacity that could carry some of its tokens. Additional legal within-scale endpoints can use this capacity, provided their arithmetic-code length fits the frame. With a partial boundary scale, the receiver must also incorporate recovered tokens into both residual reconstruction and finer-scale prediction context. Coordinating source-code termination, protected-frame capacity fitting and progressive reconstruction is therefore the central problem studied here.

We propose NeST-Com, a next-scale digital image communication framework for fixed per-image channel-symbol budgets. It retains scale-wise probability evaluation and selects token prefixes within scales by matching their independently terminated arithmetic-code length to the available protected-frame body capacity. Through VAR's known-token overwrite and recursive state update, received boundary tokens contribute to residual reconstruction and condition finer-scale prediction [1]. A coder-specific nondecreasing length property characterizes the budget-feasible prefix set. Shared-model comparisons and matched endpoint-granularity experiments verify gains in capacity use and reconstruction quality; receiver interventions examine the role

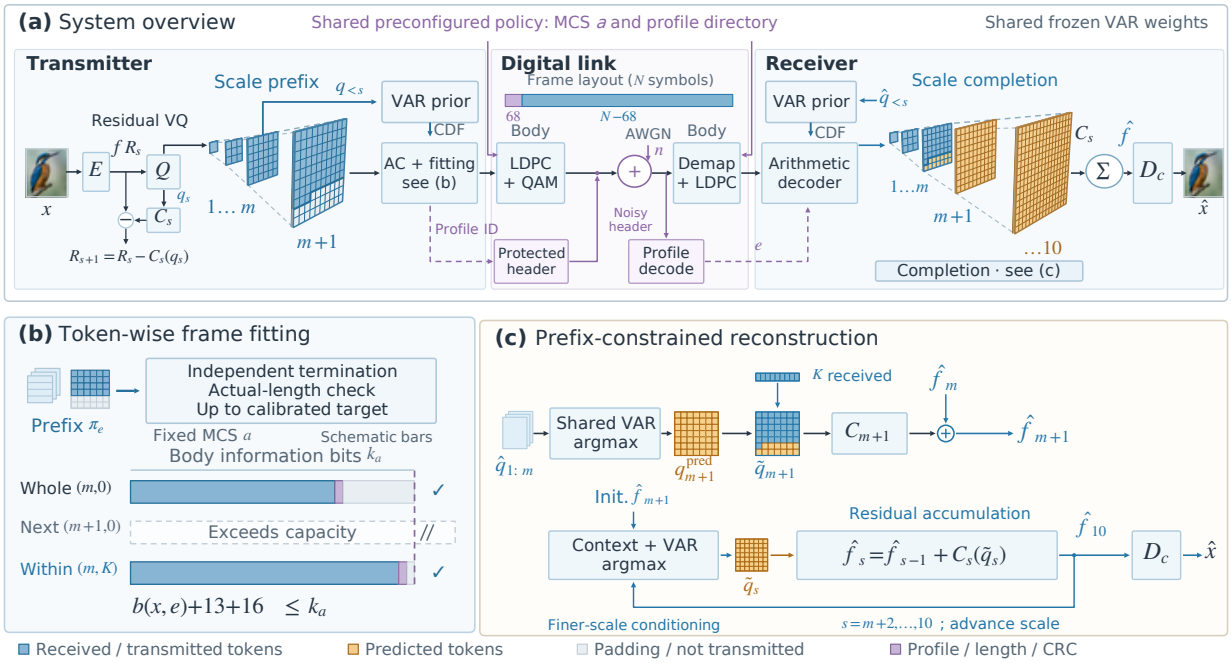


Fig. 1. NeST-Com. (a) The shared policy fixes the MCS and profile directory; the protected Profile ID selects the endpoint. VAR and known coarse tokens provide coding probabilities. (b) Independently terminated prefix bitstreams up to the target endpoint fit the body capacity; bars are schematic. (c) Received tokens overwrite boundary predictions. Recursive state updates accumulate residuals and condition each finer scale before decoding. Unreceived boundary positions retain their coarse-conditioned predictions.

of received information (Table I). System-level comparisons evaluate reconstruction quality and processing cost across communication schemes.

II. Proposed NeST-Com Framework

A. Next-Scale Representation and Digital Frame

We transmit $x \in [0,1]^{256\times256\times3}$ using N complex symbols over AWGN,

$$y = s + n, \quad n \sim \mathcal{CN}(0, \sigma^2 I_N), \quad \mathrm{SNR} = 10\log_{10}(E_s/\sigma^2), \tag{1}$$

with nominal $E_s = 2$. TX uses the frozen policy for its design SNR; RX uses the actual noise variance for LLRs. Digital constellations have average energy two without per-frame renormalization. A 12-bit profile ID, CRC16, and terminated rate-1/2 convolutional code occupy 68 rotated-BPSK symbols $\{\pm(1+j)\}$. Under a preconfigured policy, both terminals know the MCS and profile directory; the 12-bit profile identifies the endpoint within that directory. The remaining $N - 68$ symbols carry the LDPC-coded body, with public frame-dependent scrambling after rate matching and inverse scrambling before decoding. Continuous JSCC data are normalized per image to energy two; required side information is paid within N.

The frozen VAR tokenizer produces ten residual token maps q_s, with side lengths $(1, 2, 3, 4, 5, 6, 8, 10, 13, 16)$ and a 4096-entry codebook [1]. Write L_s for the number of positions and C_s for codebook lookup, resizing to the full latent grid, and the scale-dependent residual transform. The latent sum $\sum_s C_s(q_s)$ progressively refines a full-field description. Decoder D_c is adapted on 20 000 ImageNet training images using MSE plus 0.1 LPIPS, then frozen. Training mixes continuous, fully quantized and interpolated prefix-completed latents. The null-class VAR-d16 prior factorizes as

$$p_\theta(q_s \mid q_{<s}) = \prod_{i=1}^{L_s} p_\theta(q_{s,i} \mid q_{<s}). \tag{2}$$

Under the shared numerical configuration, both terminals reconstruct identical integer CDFs, with total frequency 2^{24}, for arithmetic coding [6]; no unreceived same-scale token is needed. Transmission can therefore end within a scale while retaining scale-wise probability evaluation.

B. Within-Scale Endpoints and Actual-Length Fitting

An endpoint $e = (m, K)$ retains complete coarse maps and a raster-ordered boundary prefix:

$$\pi_e = (q_1, \ldots, q_m, q_{m+1,1:K}). \tag{3}$$

For $K = 0$, including $e = (10, 0)$, the boundary prefix is empty. Raster order specifies positions without a mask. WHOLE permits $(m, 0)$ for $m = 4, \ldots, 10$. NeST permits every integer $K = 0, \ldots, L_{m+1} - 1$ for $m = 4, \ldots, 9$, together with $(10, 0)$: 651 endpoints from 30 through 680 retained tokens. Quarter/Eighth retain $K = \lfloor jL_{m+1}/d \rfloor$, $j = 1, \ldots, d-1$, for $d = 4/8$, plus complete-scale boundaries, with duplicates removed. Token is the main configuration.

For MCS $a = (M, r)$, the coded body has $n_a = (N - 68) \log_2 M$ bits and information length $k_a = \lfloor r n_a \rfloor$. Let $b(x, e)$ denote the actual bit length obtained by independently terminating the arithmetic code for token prefix π_e. Feasibility requires

$$b(x, e) + 29 \leq k_a. \tag{4}$$

The source-bit capacity utilization is $\eta(x, e) = b(x, e)/(k_a - 29)$, where $k_a - 29$ is the capacity available to the source payload. The 29 bits comprise a 13-bit length and CRC16. Remaining positions are zero-padded before 5G NR LDPC coding, rate matching [7], and QAM.

For $N = 1024$, WHOLE, NeST and Ada use the same 12 MCSs: QPSK/16-QAM/64-QAM with rates $1/2, 2/3, 3/4, 5/6$. WHOLE has seven ceilings; NeST uses 25 Quarter ceilings with Token fallback; Ada allows prefixes through 256. All 396 candidates enter a 100-image, one-realization pilot. Per method/SNR, the top three plus a preset QPSK rate-1/2 anchor advance to 1000 sources with three realizations. Maximum equal-source DINOv2 cosine [8], including gray failures, selects the policy. $N = 512$ retains its independently calibrated policies (Table IV).

A public method/MCS directory, frozen before calibration and shared across images, maps source profiles to endpoints and includes a NULL profile. One traversal shares scale-wise probability evaluation and incremental coding. Non-mutating termination produces independent prefix bitstreams, not slices of a longer codeword.

For this coder's fixed token/CDF traversal and non-mutating finish, $b_t = E_t + P_t + 2$ after t tokens, where E_t and P_t count emitted and pending bits. Renormalization increments $E_t + P_t$; other interval updates preserve it. Thus b_t is nondecreasing, and feasible endpoints form an initial segment for fixed MCS and ceiling. TX selects the longest feasible token prefix. If no source prefix fits, TX sends a NULL frame with a two-bit zero payload, using the full N-symbol budget. The measured code scans all candidates.

C. Prefix-Constrained Reconstruction

Invalid profile CRC/catalogue, body CRC, declared length, known padding, or arithmetic parsing yields fixed gray. Valid NULL frames also produce gray; both remain in all metric denominators. Otherwise, RX decodes the actual prefix using the shared CDFs. At $e = (10, 0)$, all maps are received and directly decoded. For an incomplete prefix, RX applies VAR's known-token overwrite and recursive state update [1]. It predicts the boundary map by argmax conditioned on complete received maps, then overwrites received positions:

$$\tilde{q}_{m+1,i} = \begin{cases} \hat{q}_{m+1,i}, & i \le K, \\ q^{\text{pred}}_{m+1,i}, & i > K. \end{cases} \tag{5}$$

Received boundary tokens serve two roles: they correct the residual representation and, through the merged boundary map, condition finer-scale prediction. Unreceived boundary positions retain their original coarse-conditioned prediction; there is no same-scale prediction loop. Complete received maps remain unchanged. After generating the finer maps,

$$\hat{x} = D_c\left(\sum_{s=1}^{10} C_s(\tilde{q}_s)\right). \tag{6}$$

At the last scale, overwrite only changes the residual contribution. Supplement S1 gives the full algorithm, coder-specific proof, model configurations and numerical implementation details.

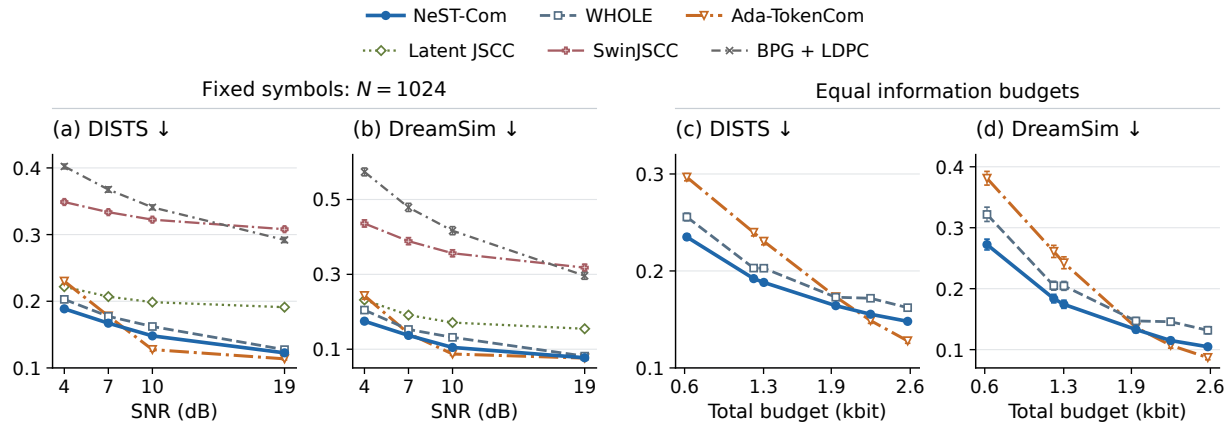


Fig. 2. Source-referenced reconstruction on 500 ImageNet images. (a,b) Six methods at $N = 1024$; WHOLE, NeST and Ada share the 12-MCS calibration procedure. Three outcomes per source include gray failures. (c,d) Equal, error-free information budgets pay 57 shared protocol bits and Ada's additional 10-bit class; WHOLE/NeST share ceiling $(10, 0)$. Points and bars are equal-source means and pointwise 95% source-bootstrap intervals after within-source averaging. Ada-TokenCom is the predicted-class, fixed-frame adaptation. Lower values are favorable.

III. Experimental Results

A. Experimental Setup

We test 500 ImageNet validation images [9], disjoint from 1000 calibration images, at 256×256: $N = 1024$ at 4/7/10/19 dB and $N = 512$ at 4/10/19 dB. We average three realizations per image, including gray failures, then compute equal-source means and direct paired 95% intervals using 10 000 source resamples. Controls use a fixed 100-image subset. Calibrated token systems and matched controls share noise; external systems retain their original noise.

Latent JSCC maps tokenizer latents to symbols and shares D_c (MSE, LPIPS and latent-consistency losses, 100k updates). SwinJSCC [10] uses RGB MSE (200k updates, calibration-MSE selection). Both train on 20 000 images at $N = 1024$. BPG [11] uses LDPC/QAM. Ada-TokenCom [3] is a predicted-class, fixed-frame adaptation: LlamaGen-L/VQ16, prefix arithmetic

coding, CFG=4, $T = 1$, top-$k = 2000$, top-$p = 1$, and a paid 10-bit ConvNeXt class. Long-term control and ARQ are excluded.

DISTS [12], DreamSim [13], PSNR, LPIPS-Alex [14] and CLIP-I [15] use native float RGB with fixed preprocessing. Timing uses 32 images, two warmups, three measurements, batch one, one RTX PRO 6000 and two physical CPU cores. We separately measure GPU-synchronized TX (RGB-to-symbols) and RX (symbols-to-RGB) processing times. They include coding, link processing and generation, excluding loading, I/O and scoring; total processing time is their sum. Supplement S2–S8 gives complete results, controls, examples, timing and transfer details, including ARPC's specified digital-link adaptation.

B. Within-Scale Fitting and Granularity

NeST and WHOLE share model, coder and receiver with family-specific policies (Table I(A)). All five paired metrics favor NeST at seven budget/SNR points. At $N = 1024$/10 dB, utilization rises from 77.70% to 99.79%, DISTS falls from 0.1621 to 0.1481, and DreamSim from 0.1317 to 0.1048. PSNR gains are 0.49–0.88 dB across $N = 1024$ points.

Block B isolates granularity by fixing model, 16-QAM rate 2/3, ceiling $(8, 126)$, fitting rule and noise. Eighth retains approximately 92% of the Whole-to-Token gain in PSNR, DISTS and DreamSim. Token raises utilization from 97.44% to 99.85%, adding $0.066\,[0.034, 0.101]$ dB PSNR, with DISTS/DreamSim differences $-0.0011\,[-0.0020, -0.0002]$/$-0.0022\,[-0.0042, -0.0001]$, at 30.7 ms (14.1%) additional TX cost. Eighth offers lower cost; Token prioritizes fuller capacity use and the additional measured quality gain.

The $N = 512$/10-dB control fixes 100 images, three shared noise realizations, 16-QAM rate 2/3, ceiling $(7, 0)$, model and receiver. Eighth retains 87% of the Whole-to-Token PSNR gain;

TABLE I

WITHIN-SCALE TRANSMISSION. (A) SHARED-MODEL NEST–WHOLE DIFFERENCES. (B) MATCHED ENDPOINT GRANULARITY. (C) ORDERED RECEIVER INTERVENTIONS. BRACKETS ARE SOURCE-PAIRED 95% INTERVALS; SAMPLE SIZES ARE SPECIFIED FOR EACH BLOCK.

A. 500 sources, $N = 1024$, uniformly calibrated policies

SNR	Util. W→N (%)	ΔDISTS	ΔDreamSim
4	73.81→99.56	-0.0142 [-0.0155, -0.0127]	-0.0297 [-0.0330, -0.0264]
7	83.19→99.71	-0.0103 [-0.0139, -0.0068]	-0.0159 [-0.0207, -0.0112]
10	77.70→99.79	-0.0141 [-0.0153, -0.0128]	-0.0270 [-0.0299, -0.0241]
19	85.24→99.83	-0.0048 [-0.0053, -0.0043]	-0.0043 [-0.0051, -0.0036]

B. 100 quality sources; 32 timing sources; ceiling $(8, 126)$

Grid	Util. (%)	DISTS	DreamSim	TX (ms)
Whole	78.13	0.1650	0.1387	185.2
Quarter	95.19	0.1535	0.1143	215.8
Eighth	97.44	0.1516	0.1137	217.8
Token	99.85	0.1505	0.1115	248.4

C. 100 sources, 10 dB, same received prefixes

Intervention	ΔPSNR (dB)	ΔDISTS	ΔDreamSim
B0–A0	+0.287 [0.240, 0.335]	-0.0062 [-0.0077, -0.0048]	-0.0113 [-0.0147, -0.0083]
C0–B0	+0.537 [0.404, 0.682]	-0.0083 [-0.0102, -0.0065]	-0.0159 [-0.0205, -0.0116]

A0: prior completion without received boundary tokens. B0: A0 plus boundary residual correction. C0: B0 plus updated finer-scale conditioning.



Fig. 3. Selected reconstructions at $N = 1024$ and 10 dB. WHOLE, NeST and the predicted-class fixed-frame Ada-TokenCom adaptation use the policies in Table IV. Values are full-image DISTS↓ / DreamSim↓ for each method's first prescribed realization. The sources were selected for favorable perceptual examples before recalibration.

Token further improves PSNR, LPIPS and DreamSim, with DISTS/CLIP-I intervals crossing zero.

Block C uses identical received NeST prefixes. A0 generates boundary/finer maps without received boundary tokens; B0 adds their residual correction with A0's finer-scale conditioning; C0 also updates that conditioning. At 10 dB, B0–A0/C0–B0 add 0.287/0.537 dB PSNR, with paired PSNR/DISTS/DreamSim intervals favoring both stages.

C. Perceptual Reconstruction and Processing Cost

Fig. 2(a,b) compares six methods at equal symbol budgets. WHOLE shares NeST's model; external systems differ in representation, coding and receiver. NeST has lower DISTS/DreamSim than SwinJSCC, Latent JSCC and BPG at all four $N = 1024$ SNRs (24 paired intervals excluding zero). SwinJSCC and BPG have higher PSNR.

TABLE II

AUXILIARY QUALITY ON 500 TEST IMAGES AT $N = 1024$, 10 DB. MEANS INCLUDE THREE REALIZATIONS PER SOURCE AND GRAY FAILURES.

Method	PSNR (dB)↑	LPIPS↓	CLIP-I↑
WHOLE	19.31	0.199	0.858
NeST-Com	20.19	0.166	0.877
Ada-TokenCom	19.56	0.167	0.884
Latent JSCC	20.80	0.173	0.789
SwinJSCC	23.03	0.469	0.669
BPG + LDPC	22.54	0.589	0.635

Against Ada, NeST improves both distances at 4 dB and DISTS at 7 dB; at 7 dB, the DreamSim difference interval crosses zero. Ada yields lower distances at 10 dB and lower DISTS at 19 dB, where the DreamSim difference interval again crosses zero. NeST has higher PSNR throughout. At 7 dB, WHOLE/NeST/Ada reject 11/7/10 of 1500 receptions each; NeST rejects one at 4 dB, with zero rejections at the other token-system points and no valid NULL frames. Table II gives auxiliary metrics at 10 dB.

Fig. 2(c,d) uses six equal information budgets (620–2577 bits), including 57 shared protocol bits plus Ada's 10-bit class. Error-free reconstruction averages three generation repeats with ceilings $(10, 0)$ for WHOLE/NeST and 256 for Ada. NeST improves all five metrics over WHOLE. Against Ada, DISTS favors NeST at the four smallest budgets and DreamSim at the first three; the 1940-bit DreamSim interval crosses zero. Ada has lower distances at 2248/2577 bits.

Under the auxiliary configurations, Ada's argmax tail on 100 fixed images worsens DreamSim at $N = 512$, 4/10 dB and $N = 1024$, 10 dB relative to sampling.

Fig. 3 illustrates NeST preserving truck markings and foreground contours. These selected cases favor NeST in DISTS/DreamSim and SwinJSCC in PSNR.

Timing uses earlier profile directories with the current $N = 1024$/10-dB MCS and source-prefix ceilings. Measured TX/RX is 248.44/403.99 ms for NeST and 215.29/366.42 ms for WHOLE. Totals are 652.43 ms for NeST and 39.17 ms for SwinJSCC. Compiled Ada's 3.064 s is paired with its own 32-image quality; main quality uses the original implementation.

D. Reduced Budget and Transfer to Kodak

At $N = 512$, all five NeST–WHOLE intervals favor NeST, with PSNR gains of 0.50–0.99 dB. At 10 dB, both use ceiling $(7, 0)$ and 16-QAM rate 2/3; utilization rises from 79.30% to 99.54%. NeST has higher PSNR and lower LPIPS than Ada throughout, and lower DISTS/DreamSim at 4/10 dB. Ada has lower distances at 19 dB; both paired intervals exclude zero.

Kodak's 24 center crops reuse the original ImageNet-calibrated policies. NeST improves PSNR/DISTS/DreamSim over WHOLE at all three SNRs; all 864 receptions succeed. NeST has lower distances than Ada at 4 dB; Ada has lower DISTS at 10/19 dB, with DreamSim intervals crossing zero. Historical Ada uses 64-QAM rate 5/6 at 19 dB, versus the current 16-QAM rate 5/6.

E. Operation Near the Channel Threshold

With fixed 10-dB timing policies and actual noise variance at RX, 100 images and 20 noise realizations yield 30 000 receptions over 8–10 dB. All methods reject every frame at 8 dB. WHOLE/NeST/Ada rejection rates are 15.75/14.95/13.50% at 9 dB, 0.10–0.20% at 9.5 dB and zero at 10 dB. Metrics retain all rejected gray outputs.

TABLE III

REDUCED-BUDGET RECONSTRUCTION ON 500 SOURCES AT $N = 512$. MEANS INCLUDE ALL FAILURES; PSNR DIFFERENCES USE DIRECT SOURCE-PAIRED 95% INTERVALS.

SNR / method	PSNR↑	DISTS↓	DreamSim↓
4 / WHOLE	15.11	0.2580	0.3247
4 / NeST-Com	15.61	0.2363	0.2740
4 / Ada-TokenCom	11.32	0.2996	0.3847
NeST–WHOLE ΔPSNR: +0.504 [0.440, 0.573] dB			
10 / WHOLE	16.82	0.2030	0.2049
10 / NeST-Com	17.41	0.1922	0.1837
10 / Ada-TokenCom	13.16	0.2367	0.2582
NeST–WHOLE ΔPSNR: +0.592 [0.544, 0.643] dB			
19 / WHOLE	18.64	0.1718	0.1458
19 / NeST-Com	19.63	0.1554	0.1150
19 / Ada-TokenCom	17.69	0.1496	0.1078
NeST–WHOLE ΔPSNR: +0.994 [0.940, 1.048] dB			

IV. CONCLUSION

NeST-Com fits within-scale prefixes by arithmetic-code length to carry more source information in fixed protected frames. Shared-model comparisons and matched controls link fuller capacity use to better reconstruction. Eighth and Token offer quality–computation choices, with finer termination yielding additional measured gains at higher TX cost. System-level comparisons show Ada's perceptual advantages at some operating points and SwinJSCC's substantially faster processing.

TABLE IV

FROZEN POLICIES: MCS ENTRIES ARE (BITS/SYMBOL, CODE RATE); e IS THE TARGET CEILING, WITH FEASIBLE FALLBACK. ADA'S CEILING IS 256 TOKENS. BUDGETS ARE CALIBRATED INDEPENDENTLY.

N/SNR	a_{W}	e_{W}	a_{N}	e_{N}	a_{Ada}
1024/4	$2, 2/3$	$(7, 0)$	$2, 2/3$	$(7, 25)$	$2, 2/3$
1024/7	$4, 1/2$	$(8, 0)$	$4, 1/2$	$(8, 0)$	$4, 1/2$
1024/10	$4, 2/3$	$(9, 0)$	$4, 2/3$	$(8, 126)$	$4, 2/3$
1024/19	$6, 5/6$	$(10, 0)$	$6, 5/6$	$(10, 0)$	$4, 5/6$
512/4	$2, 2/3$	$(6, 0)$	$2, 2/3$	$(5, 27)$	$2, 2/3$
512/10	$4, 2/3$	$(7, 0)$	$4, 2/3$	$(7, 0)$	$4, 2/3$
512/19	$6, 5/6$	$(8, 0)$	$6, 5/6$	$(8, 84)$	$6, 5/6$

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